

**UIL HIGH SCHOOL SCIENCE CONTEST
ANSWER KEY
2025 INVITATIONAL B**

Biology

B01. A
B02. E
B03. D
B04. D
B05. B
B06. E
B07. E
B08. A
B09. C
B10. A
B11. C
B12. B
B13. B
B14. D
B15. B
B16. C
B17. A
B18. E
B19. C
B20. D

Chemistry

C01. B
C02. E
C03. C
C04. D
C05. E
C06. C
C07. C
C08. A
C09. A
C10. B
C11. D
C12. A
C13. A
C14. B
C15. D
C16. B
C17. E
C18. B
C19. C
C20. E

Physics

P01. E
P02. A
P03. D
P04. A
P05. B
P06. C
P07. D
P08. C
P09. A
P10. D
P11. C
P12. E
P13. B
P14. A
P15. E
P16. E
P17. A
P18. B
P19. C
P20. D

CHEMISTRY SOLUTIONS – UIL INVITATIONAL B 2025

- C01. (B) Aluminum nitrate is $\text{Al}(\text{NO}_3)_3$. The molar mass is $26.98 + 3 \times 14.01 + 9 \times 16.00 = 213.01 \text{ g/mol}$
- C02. (E) $2 \text{CH}_3\text{OH}(\ell) + 3 \text{O}_2(g) \rightarrow 2 \text{CO}_2(g) + 4 \text{H}_2\text{O}(g)$
- C03. (C) The electron has moved to a higher energy level so the atom must have absorbed some energy from its surroundings.
- C04. (D) In valence bond theory the s and p orbitals that hybridize must be in the same energy level (and any d orbitals that are involved are in $n - 1$). Since there is only one s orbital in each energy level you can never have two s orbitals hybridize together.
- C05. (E) The prefixes *mono-*, *di-*, and so on are not used when naming ionic compounds, so B and C are wrong. When a metal only has one ionic valence state, such as Zn^{2+} or Ag^+ , the oxidation state is not included in the name, so A and D are simply zinc nitrate and silver bromide respectively. Lead has two ionic oxidation states so the oxidation state must be specified in the compound name, and the iodate ion is IO_3^- .
- C06. (C) $P_1V_1 = P_2V_2$, so $V_2 = P_1V_1/P_2$ $P_1 = 9.0 \text{ atm}$, $V_1 = 5.0 \text{ L}$, $P_2 = 3.0 \text{ atm}$ $V_2 = (9.0)(5.0)/(3.0) = 15 \text{ L}$. V_2 is the volume at the end which is the volume of both bulbs combined, so the volume of the bulb on the right is $15 \text{ L} - 5.0 \text{ L} = 10 \text{ L}$
- C07. (C) Only ethanol and methanol have hydrogen bonding, and ethanol has a higher molar mass than methanol (which means it is more polarizable and has more dispersion forces), so the intermolecular forces will be strongest in ethanol, giving it the highest boiling point among these compounds.
- C08. (A) $3000\text{J}/4200 \text{ J/mol} = 0.7143 \text{ moles}$. $0.7143 \text{ moles} \times 140.22 \text{ g/mol} = 100.157 \text{ grams}$
- C09. (A) $-m_{\text{MCM}}\Delta T_{\text{M}} = m_{\text{water}}c_{\text{water}}\Delta T_{\text{water}}$
 $c_{\text{M}} = ? \text{ J/g}^\circ\text{C}$
 $\Delta T_{\text{M}} = (27.6 - 88.0) = -60.4 \text{ }^\circ\text{C}$
 $m_{\text{water}} = 125 \text{ g}$
 $c_{\text{water}} = 4.184 \text{ J/g}^\circ\text{C}$
 $\Delta T_{\text{water}} = (27.6 - 25.0) = 2.6 \text{ }^\circ\text{C}$

$$c_{\text{M}} = \frac{-m_{\text{w}}c_{\text{w}}\Delta T_{\text{w}}}{m_{\text{M}}\Delta T_{\text{M}}} = \frac{-(125)(4.184)(2.6)}{(50.14)(-60.4)} = 0.449 \text{ J/g}^\circ\text{C}$$

On the data sheet the metal with a specific heat of $0.449 \text{ J/g}^\circ\text{C}$ is chromium.

- C10. (B) Z is a solid and is not part of the K_{eq} calculation.

$$K_{\text{eq}} = \frac{P_{\text{Y}}}{P_{\text{W}}^2 P_{\text{X}}} = \frac{(1.8)}{(0.75)^2 (0.85)} = 3.76$$

- C11. (D) moles of $\text{Ba}(\text{OH})_2 = 0.100 \text{ L} \times 0.10 \text{ M} = 0.010 \text{ moles Ba}(\text{OH})_2$ moles of $\text{OH}^- = 2 \times$
 moles of $\text{Ba}(\text{OH})_2 = 0.020 \text{ moles}$. $[\text{OH}^-] = 0.020 \text{ mol} / 0.300 \text{ L} = 0.0667 \text{ M}$.
 $\text{pOH} = 1.176$. $\text{pH} = 14 - \text{pOH} = 12.82$

- C12. (A) The precipitate is lead(II) chloride, PbCl_2 , so precipitation will begin when
 $[\text{Pb}^{2+}][\text{Cl}^-]^2 = 1.7 \times 10^{-5}$. $[\text{Cl}^-]^2 = K_{\text{sp}}/[\text{Pb}^{2+}] = 1.7 \times 10^{-5}/0.0010 = 0.0170$, so
 $[\text{Cl}^-] = (0.0170)^{1/2} = 0.13 \text{ M}$
- C13. (A) The oxidation state of oxygen in a compound is almost always -2 (except in peroxides or OF_2), and since CO is neutral the oxidation number on the carbon atom must be $+2$.
- C14. (B) (I hope you knew there would be a catalyst question on Invitational B after the hint in the answer to question 20 on Invitational A.)
- C15. (D) The vaporization temperature and pressure are irrelevant to solving this problem. The problem gives you a volume of liquid methanol and wants to know how many molecules there are in it. Use the density to convert the volume to mass, then divide by molar mass to get moles.
 $325 \text{ mL} \times 0.792 \text{ g/mL} = 257.4 \text{ grams}$. $257.4 \text{ g} / 32.05 \text{ g/mol} = 8.03 \text{ moles of CH}_3\text{OH}$.
 $8.03 \text{ moles} \times 6.02 \times 10^{23} \text{ molecules/mol} = 4.83 \times 10^{24} \text{ molecules}$.
- C16. (B). Only statement II is correct. Heat is a form of energy, but temperature is a measure of the average kinetic energy of the particles. During a phase change (melting or vaporization) it is possible to add heat without changing the temperature of the substance because the heat energy that is added is used to overcome intermolecular attractions between the particles. Heat is transferred to the ice and the water gets colder because it loses heat. There is no such thing as “cold” that can be transferred from one thing to another. Heat does not have mass. *Heat* does not rise, less dense objects rise. Heat causes materials to expand and become less dense.
- C17. (E) $1 \text{ C}_3\text{H}_8 + 5 \text{ O}_2 \rightarrow 3 \text{ CO}_2 + 4 \text{ H}_2\text{O}$ I didn’t give you the products of this reaction because I assume you learned them from the question on Invitational A if you didn’t know them already. Question 2 on this test also provided a reminder.
- C18. (B) You don’t have to actually solve this problem to get the answer because you know the mixture concentration has to be between 0.95 M and 1.15 M . That leaves only answers B, C, and D. Since you have more of the 0.95 M solution than you have of the 1.15 M solution, the mixture concentration has to be closer to 0.95 M than it is to 1.15 M . That means it has to be answer B. But you should solve the problem anyway just in case this same lab partner makes a more complicated mistake on a later test using different solutions where the answer can’t be reasoned out quite this easily...
Moles of chloride in the $\text{NaCl} = (0.310 \text{ L})(1.15 \text{ M}) = 0.3565 \text{ mol}$.
Moles of chloride in the $\text{KCl} = (0.925 \text{ L})(0.95 \text{ M}) = 0.8788 \text{ mol}$. Total moles = 1.235 moles Cl^- .
Total volume = $0.310 \text{ L} + 0.925 \text{ L} = 1.235 \text{ L}$. $1.235 \text{ mol}/1.235 \text{ L} = 1.00 \text{ M}$
- C19. (C)
- C20. (E) $2 \text{ Al(s)} + 6 \text{ HF(g)} \rightarrow 2 \text{ AlF}_3\text{(s)} + 3 \text{ H}_2\text{(g)}$ You don’t actually need a balanced equation for the reaction as long as you realize you need three moles of HF for every mole of Al .
 $165 \text{ g Al} / 26.98 \text{ g/mol} = 6.1156 \text{ mol Al}$, so you need $3 \times$ that many moles of $\text{HF} = 18.3469$ moles HF . $V = nRT/P$ so at STP $V = (18.3469)(0.08216)(273)/1 = 411.0 \text{ L of HF gas}$

PHYSICS SOLUTIONS – UIL INVITATIONAL B 2025

- P01. (E) page 52: “The crisis in physics precipitated by Maxwell’s Equations and the Michaelson-Morley experiment forced physicists to reconsider some deeply held notions. The most fundamental of these notions was the idea of universal time.”
- P02. (A) page 69: “The most dramatic example of the clock-slowness effects of relativity involves exotic particles known as muons.”
- P03. (D) page 81: “The problem of length measurement is really a problem of synchronization....”
- P04. (A) The ice giants, Uranus and Neptune, formed from the accretion of solid icy material in the outer solar system. The icy bodies that now form the mantles of the ice giants are primarily composed of water, ammonia, and methane.
- P05. (B) The total magnitude of the electric field is the length of the hypotenuse of a triangle where the horizontal leg is 140.0N/C in length and the angle of the hypotenuse with respect to horizontal is 35.0° . Based on the triangle, we surmise $\cos \theta = \frac{x}{h} \rightarrow \cos(35.0^\circ) = \frac{140.0}{h} \rightarrow h = \frac{140.0}{\cos(35.0^\circ)} = 171\text{N/C}$. This is the total magnitude of the electric field.
- P06. (C) This problem has two parts: in the first part, the truck is accelerating, and in the second part, the truck is moving at constant speed. During the first part, we know that the truck starts from rest, ends with a speed of 30.0m/s, and accelerates at 2.25m/s^2 . We do not know for how long the truck accelerated, so we’ll use the equation $v_f^2 = v_i^2 + 2a\Delta x_1 \rightarrow (30.0)^2 = (0)^2 + 2(2.25)\Delta x_1$. This gives $900 = 4.50\Delta x_1 \rightarrow \Delta x_1 = 200.0\text{m}$. This is the distance traveled during the first part. For the second part, the truck travels at a constant speed of 30.0m/s for 5.00seconds. For this part, we know the acceleration is zero. We will use $\Delta x_2 = vt + \frac{1}{2}at^2 = (30.0)(5.00) + (0.5)(0)(5)^2 = 150.0\text{m}$. Therefore, the total distance traveled by the truck is $d = \Delta x_1 + \Delta x_2 = 200.0 + 150.0 = 350.0\text{m}$
- P07. (D) First, we need to construct a free-body force diagram. Because there is no friction, we only have three forces acting on the crate: gravity (mg , downward), the normal force (F_N , upward), and the pulling force (F , up and right). The pulling force is at an angle, so we will break it into its components. In the horizontal, we have $F_h = F \cos \theta = (120.0) \cos(31.0^\circ) = 102.9\text{N}$. In the vertical, we have $F_v = F \sin \theta = (120.0) \sin(31.0^\circ) = 61.8\text{N}$. Now we sum our forces. The motion is entirely horizontal, so the vertical forces must sum to zero. For the vertical, $\sum F_y = F_N + F_v - mg = 0$. This gives $F_N + 61.8 - (65)(9.8) = 0 \rightarrow F_N = 575\text{N}$. This is nice to know, but useless for solving the problem. The acceleration is horizontal, so summing the horizontal forces gives $\sum F_x = F_h = ma$. This gives $102.9 = (65)a \rightarrow a = 1.58\text{m/s}^2$, which is the answer.
- P08. (C) Again, we start with a free body force diagram. There is no friction, so we only have three forces: gravity (mg , downward), the normal force (F_N , upward), and the pulling force (F , up and right). As in problem P07, we break the pulling force into its components. In the horizontal, we have $F_h = F \cos \theta = (40.0) \cos(31.0^\circ) = 34.3\text{N}$. In the vertical, we have $F_v = F \sin \theta = (40.0) \sin(31.0^\circ) = 20.6\text{N}$. The motion of the crate is entirely horizontal, so it is only the horizontal forces that are needed to calculate the work. The only horizontal force is the horizontal component of the applied force: $\sum F_x = F_h = 34.3\text{N}$. Work is defined as force multiplied by distance, so for the horizontal motion of the crate, we have $W = F_h d_h = (34.3\text{N})(45.0\text{m}) = 1543\text{J}$.

- P09. (A) We first consider the force diagram for this situation. There are two forces acting on the ball: gravity (mg , directed downward) and tension (T , directed upward along the rope). Because the tension acts at an angle, it must be broken into components. In the vertical, we have $T_y = T \cos \theta$ and in the horizontal, we have $T_x = T \sin \theta$. The trigonometric functions look reversed, but that is because we are working with an angle measured with respect to the vertical. Now, since the ball's motion is entirely horizontal, there is no acceleration in the vertical direction. Thus, the forces in the vertical must sum to zero: $\sum F_y = T \cos \theta - mg = 0 \rightarrow T \cos(38.0) = (3.40)(9.80) \rightarrow T = 42.3\text{N}$. This gives us the tension in the rope.
- In the horizontal direction we have only a single force component: $\sum F_x = T \sin \theta = (42.3) \sin(38.0) = 26.0\text{N}$. This horizontal force provides the centripetal acceleration needed to keep the ball moving in a circle. Thus, $\sum F_x = 26.0 = mr\omega^2$. The last piece we need is the radius of the circle traced out by the ball. The radius equals the distance from the pole to the ball, which can be found using the length of the rope: $r = L \sin \theta = (1.50) \sin(38.0) = 0.9235\text{m}$. Putting it all together, we obtain $26.0\text{N} = (3.40\text{kg})(0.9235\text{m})\omega^2 \rightarrow \omega^2 = 8.28 \rightarrow \omega = 2.88\text{rad/s}$.
- P10. (D) The equation relating the length of a stationary pendulum to its period is $T = 2\pi\sqrt{\frac{L}{g}}$. Using the initial period, we can find the length of your pendulum: $0.750\text{s} = 2\pi\sqrt{\frac{L}{9.80}} \rightarrow \frac{L}{9.80} = \left(\frac{0.750}{2\pi}\right)^2 \rightarrow L = 0.1396\text{m}$. On the elevator, the length of your pendulum remains the same, but the acceleration has changed. This can be included in the equation: $T' = 2\pi\sqrt{\frac{L}{g+a}}$ where a is the acceleration of the elevator.
- Putting in the known and calculated values, we get: $0.650\text{s} = 2\pi\sqrt{\frac{0.1396\text{m}}{9.80+a}} \rightarrow \frac{0.1396}{9.80+a} = \left(\frac{0.650}{2\pi}\right)^2 \rightarrow 0.1396 = 0.0107(9.80 + a)$. This leads to $0.1396 = 0.1049 + 0.0107a \rightarrow 0.0107a = 0.0347 \rightarrow a = 3.25\text{m/s}^2$.
- P11. (C) This is a simple calorimetry problem without any phase changes. We start with: $\sum Q = 0$, which leads to $(mc\Delta T)_{\text{water}} + (mc\Delta T)_{\text{aluminum}} = 0$. Putting in the known values, and equating the changes in temperature to $\Delta T = T_{\text{final}} - T_{\text{initial}}$, we get $(0.100\text{kg})(4186)(T_{\text{final}} - T_{\text{initial}})_{\text{water}} + (0.020\text{kg})(900)(T_{\text{final}} - T_{\text{initial}})_{\text{aluminum}} = 0$.
- Recalling that differences in temperature in Celsius are the same as differences in Kelvin, and noting that the final temperature of the aluminum equals the final temperature of the water, we get: $(418.6)(55.0^\circ\text{C} - 22.0^\circ\text{C}) + (18.0)(55.0^\circ\text{C} - T_{\text{initial,Al}}) = 0$.
- This leads to $13814 + 990 - 18T_{\text{initial,Al}} = 0 \rightarrow 18T_{\text{initial,Al}} = 14804 \rightarrow T_{\text{initial,Al}} = 822^\circ\text{C}$.
- P12. (E) We use the power dissipated to find the current flowing in the $1.20\text{k}\Omega$ resistor: $P = I^2R \rightarrow 0.2016 = I^2(1200) \rightarrow I = 0.01296\text{A}$. Since the two resistors are connected in series, we know that this same current also flows through the $1.50\text{k}\Omega$ resistor. Using Ohm's Law, we can now find the voltage across each resistor: $V_1 = IR_1 = (0.01296)(1200) = 15.55\text{V}$, and $V_2 = IR_2 = (0.01296)(1500) = 19.44\text{V}$. For a series circuit, the total voltage equals the sum of the individual voltages.
- Thus, $V = V_1 + V_2 = 15.55 + 19.44 = 34.99 \approx 35.0\text{V}$ is the voltage produced by the power supply.
- P13. (B) The charge Q_2 starts a long distance away, so the initial electric potential energy associated with these two charges is zero ($U_i = 0$). Once Q_2 is brought to the 20.0cm mark, the energy associated with the two charges is found using $U_f = \frac{kQ_1Q_2}{r}$. This gives $U_f = \frac{(8.99 \times 10^9)(4.50 \times 10^{-6})(1.25 \times 10^{-6})}{(20.0\text{cm} - 5.00\text{cm})} = \frac{0.05057\text{Jm}}{0.15\text{m}} = 0.337\text{J}$. Thus, the total work needed to bring charge Q_2 to the 20.0cm mark is $W = U_f - U_i = 0.337 - 0 = 0.337\text{J}$.

- P14. (A) There are three types of magnetic materials: *diamagnetic*, *paramagnetic*, and *ferromagnetic*. Dipoles in a *diamagnetic* material will anti-align with an external magnetic field but will not stay anti-aligned when the external field is removed (choice B). Dipoles in a *ferromagnetic* material will align with an external magnetic field and will remain aligned even after the external field is gone (choice C). The dipoles in a *paramagnetic* material like aluminum will align with an external magnetic field but will not remain aligned when the external field is removed (choice A, the correct answer).
- P15. (E) The light ray doesn't change direction at the first surface since it intersects the surface along the normal line (that is perpendicular to the surface). Thus, we can ignore the first surface. At the second surface, the ray intersects at 45° to the surface. This means that the angle with respect to the normal line is also 45° . Thus, the incident angle is $\theta_i = 45^\circ$. The angle at which the light ray exits the prism is 23° with respect to the surface. This gives an angle with respect to the normal line of $\theta_r = 90 - 23 = 67^\circ$, which is the refracted angle. With the angles determined, and noting that the index of refraction of air is 1.00, we can use Snell's Law to find the index of refraction of the prism: $n_1 \sin \theta_i = n_2 \sin \theta_r \rightarrow n \sin(45.0) = (1.00) \sin(67.0) \rightarrow 0.7071n = 0.9205 \rightarrow n = 1.30$.
- P16. (E) Since we are given the size of the snail and the size of the image of the snail, we can calculate the magnification: $M = \frac{h'}{h} = \frac{3.75\text{mm}}{1.22\text{mm}} = 3.07$. The image is upright, which confirms that the magnification is positive. We also know the object location, $p = 6.00\text{cm}$. We can go to the magnification formula to find the image location: $M = -\frac{q}{p} \rightarrow 3.07 = -\frac{q}{6.00} \rightarrow q = -18.44\text{cm}$. The image location is negative, which means that this image is virtual. Finally, we can determine the focal length of the lens by using $\frac{1}{p} + \frac{1}{q} = \frac{1}{f} \rightarrow \frac{1}{6.00} + \frac{1}{-18.44} = \frac{1}{f} \rightarrow f = 8.89\text{cm}$.
- P17. (A) The wavelengths of emitted photons are based on the energy differences between the various levels. With four energy levels, this element can produce up to six different wavelengths. The six energy differences are $\Delta E_1 = 2.82 - 0.00 = 2.82\text{eV}$, $\Delta E_2 = 2.07 - 0.00 = 2.07\text{eV}$, $\Delta E_3 = 1.63 - 0.00 = 1.63\text{eV}$, $\Delta E_4 = 2.82 - 1.63 = 1.19\text{eV}$, $\Delta E_5 = 2.82 - 2.07 = 0.75\text{eV}$, $\Delta E_6 = 2.07 - 1.63 = 0.44\text{eV}$. This gives emitted wavelengths of $\lambda_1 = \frac{1240\text{eVnm}}{2.82\text{eV}} = 440\text{nm}$, $\lambda_2 = \frac{1240\text{eVnm}}{2.07\text{eV}} = 599\text{nm}$, $\lambda_3 = \frac{1240\text{eVnm}}{1.63\text{eV}} = 761\text{nm}$, $\lambda_4 = \frac{1240\text{eVnm}}{1.19\text{eV}} = 1042\text{nm}$, $\lambda_5 = \frac{1240\text{eVnm}}{0.75\text{eV}} = 1653\text{nm}$, $\lambda_6 = \frac{1240\text{eVnm}}{0.44\text{eV}} = 2818\text{nm}$. Thus, the wavelength that is not produced by this element is 335nm .
- P18. (B) For each half-life, the Carbon-14 content of the bone fragment decreases by a factor of two. That is, in 5730 years, the Carbon-14 content is $\frac{1}{2}$, or 50%, of its original amount. After $2 * 5730 = 11460$ years, the Carbon-14 content decreases to $\frac{1}{4}$, or 25%, of its original amount. And after $3 * 5730 = 17190$ years, the Carbon-14 content is $\frac{1}{8}$, or 12.5%, of its original amount. Thus, the bone fragment has an age of approximately three half-lives, or about 17000 years old.
- P19. (C) Since this curve is not linear, our best option is to read a data point from the plot and put those values into the formula relating the mass to the period of a mass-spring system. I'm going to use the point (100g, 0.50s). The relevant equation is $T = 2\pi \sqrt{\frac{m}{k}}$. Rearranging this equation allows us to solve for the spring constant: $\left(\frac{T}{2\pi}\right)^2 = \frac{m}{k} \rightarrow k = m \left(\frac{2\pi}{T}\right)^2$. Plugging in the values from the data point selected, we get: $k = m \left(\frac{2\pi}{T}\right)^2 = (0.100\text{kg}) \left(\frac{2\pi}{0.50\text{s}}\right)^2 = 15.8 \approx 16 \text{ N/m}$.

P20. (D) The equation describing the magnetic field strength produced at a distance r from a current-carrying wire is $|B| = \frac{\mu_0 I}{2\pi r}$. Emphasizing the relationship between the magnetic field strength $|B|$ and the inverse of the distance from the wire $\frac{1}{r}$, we see that $|B| = \left(\frac{\mu_0 I}{2\pi}\right) \frac{1}{r}$. This relationship is linear (as the plot also shows), and the slope of the line is $slope = \frac{\mu_0 I}{2\pi}$. So, we need to find the slope of the line traced out by the data. To do this, I will choose two points with which to estimate the slope. I'll use the points (5 m⁻¹, 37 μT) and (20.5 m⁻¹, 140 μT). This gives a slope of $slope = \frac{y_2 - y_1}{x_2 - x_1} = \frac{140 - 37}{20.5 - 5} = \frac{103}{15.5} = 6.65 \mu T m = 6.65 \times 10^{-6} T m$. Equating this value to the slope equation we found earlier leads to: $slope = 6.65 \times 10^{-6} T m = \frac{\mu_0 I}{2\pi} = \frac{4\pi \times 10^{-7}}{2\pi} I = (2 \times 10^{-7}) I$. This gives a current of $I = \frac{6.65 \times 10^{-6}}{2 \times 10^{-7}} = 33 \approx 35 A$.